Extra Problem Set

Intermediate Macroeconomics, EC2211

Martin Flodén, Fall 2025

To be discussed in class on Monday, October 27. Do not submit solutions!

1 U.S. debt sustainability

The average annual growth rate of nominal GDP in the United States was 4.5 percent between 2002 and 2024. The interest rate on U.S. government bonds is today around 5 percent. Suppose that these values persist in the future.

The U.S. general government debt is around 120 percent of GDP and the general government primary balance is around -3 percent of GDP.

a. How much, as a fraction of GDP, does the U.S. general government pay in interest on its debt every year?

b. Starting with debt-to-GDP at 120 percent of GDP, how high would the debt-to-GDP ratio be one year later if the primary balance remains at -3 percent of GDP.

c. How much must the primary balance adjust for the debt-to-GDP ratio to be stabilized?

2 Debt dynamics when $i < g$

The average annual growth rate of nominal GDP in the United States was 4.1 percent from 2010 to 2019. In the same time period, the average interest rate on 1-year U.S. government bonds was 0.7 percent.

Consider now the following strategy: We issue a bond B_{2025} to finance some immediate government spending. When the bond matures next year, we roll over the debt by issuing a new bond $B_{2026} = (1+i)B_{2025}$.

a. Suppose that the above interest and growth rates persist. How will our rolled over bond B_t then develop over time in relation to GDP?

b. Do you see any problems associated with this strategy?

3 Expansionary monetary policy in Europe

(Exercise 6 in Jones chapter 20) Suppose the European Central Bank decides to stimulate the European economy by reducing interest rates there. Use the AS/AD model to explain how and why this affects the U.S. economy in the short run. How does the economy return to steady state?

4 Currency crises and the demand for dollars

(Exercise 8 in Jones chapter 20) Suppose there is a currency crisis in the rest of the world, leading to an increase in demand for U.S. dollars (a "flight to safety"). Use the AS/AD framework to explain the effects of this shock on the U.S. economy. Be sure to explain carefully how and why the shock enters the AS/AD model. (*Hint:* If the rest of the world would like more dollars, what does it have to give in exchange for those dollars?)

5 The unwinding of the U.S. trade deficit

(Exercise *9 in Jones chapter 20) Suppose some shock occurs to the U.S. economy that makes foreign investors more reluctant to hold U.S. assets. Use the AS/AD framework to explain the effects of this shock on the U.S. economy. Note: There are several possible answers to this question, depending on which effect dominates. Just be clear about the case you choose to analyze.

6 Transition economies

Consider an economy that has just integrated with richer and more advanced economies. Suppose that it has a well-educated population but an old and outdated capital stock. People expect income to grow rapidly in the coming decades. (Think for example of Eastern European countries in the 1990s and early 2000s, hoping to join the European Union.)

a. According to the permanent income hypothesis, how should households' consumption and savings behavior respond to this development?

b. What is likely the marginal product of capital in this economy? How do you think investment will respond to this development?

c. What does all this imply for the current account balance in the country?

7 The time consistency problem

Consider an economy with only one time period. In the beginning of the period, households form inflation expectations π^e. After inflation expectations have been formed, the policymaker chooses the inflation level π. Finally, unemployment is realized from the Phillips curve

$$\pi = \pi^e - \phi(u - \bar{u})$$

where $\bar{u} \geq 0$ is the natural unemployment level.

Suppose that the policymaker chooses π to hold both inflation and unemployment as close to zero as possible. More precisely, she chooses π to minimize this loss function:

$$L(\pi, u) = \pi^2 + \lambda u^2$$

where λ is a parameter that indicates how much the policymaker cares about unemployment.

a. Formulate the policymaker's optimization problem.

b. Find an expression for π as a function of π^e, $\bar{u}$ and parameter values.

c. When forming expectations, households understand the policymaker's problem and anticipate her behavior. Therefore, $\pi^e = \pi$. What will then be the outcome for π, π^e and u?

d. Some have argued that monetary policy should be delegated to an independent central bank led by a governor who places little weight on unemployment. Discuss the relevance of this argument based on the results in question c).